

Generated code

```
1 def ap_sum(a,n,d):
2     total = 0
3     for i in range(n):
4         total += a + i * d
5     return total
```

Canonical solution

```
1 def ap_sum(a,n,d):
2     total = (n * (2 * a + (n - 1) * d)) / 2
3     return total
```

(a) Failure to Utilize Mathematical Optimizations from MBPP-335

Generated code

```
1 def greatest_common_divisor(a:int,b:int)->int:
2     while a != b:
3         if a > b:
4             a -= b
5         else:
6             b -= a
7     return a
```

Canonical solution

```
1 while b:
2     a, b = b, a % b
3 return a
```

(b) Inefficient/Unnecessary If-else from HumanEval-13